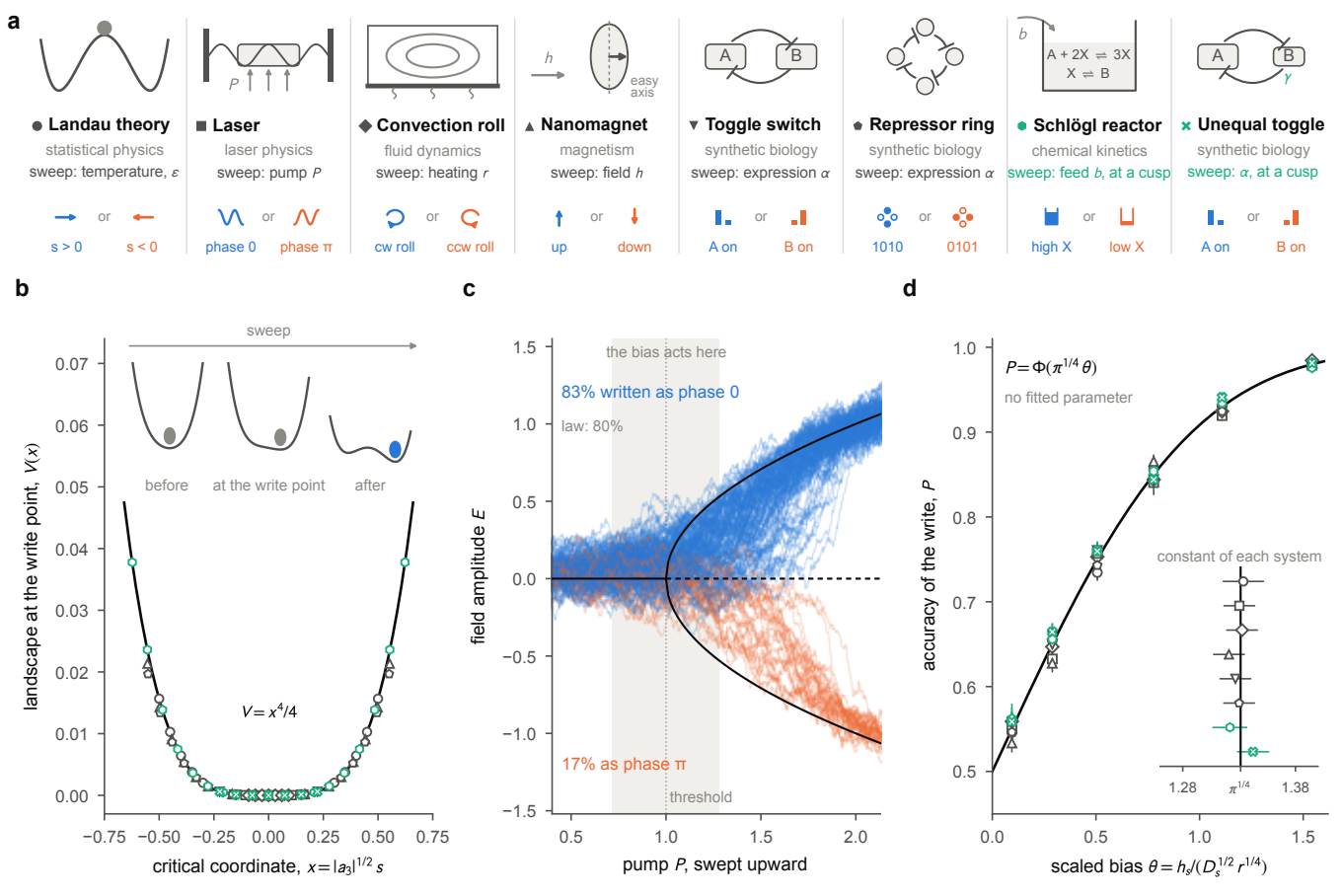

a
Landau theory
statistical physics
sweep: temperature, ε
s > 0
or
s < 0
Laser
laser physics
sweep: pump P
phase 0
phase π
Convection roll
fluid dynamics
sweep: heating r
cw roll
ccw roll
Nanomagnet
magnetism
sweep: field h
easy axis
up
down
Toggle switch
synthetic biology
sweep: expression α
A on
B on
Repressor ring
synthetic biology
sweep: expression α
1010
0101
Schlögl reactor
chemical kinetics
sweep: feed b, at a cusp
A + 2X ⇌ 3X
X ⇌ B
high X
low X
Unequal toggle
synthetic biology
sweep: α, at a cusp
A on
B on
b
sweep
before
at the write point
after
landscape at the write point, V(x)
V = x⁴/4
critical coordinate, x = |a3|^1/2 s
c
the bias acts here
83% written as phase 0
law: 80%
17% as phase π
threshold
field amplitude E
pump P, swept upward
d
P = Φ(π^1/4 θ)
no fitted parameter
constant of each system
accuracy of the write, P
scaled bias θ = hs/(Ds^1/2 r^1/4)